

Daily Science Anecdotes

Topic: All Nuclear Weapons Explode!

by Ali Hamdan

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What if we piled up all of the nuclear bombs that we currently have and pulled the trigger?

To look at the numbers, there are 15,000 nuclear warheads around the world. In more detail, the United States holds $\approx 7,000$ warheads, Russia holds a similar amount, and the remaining 1,000 are distributed across other countries. To put this into perspective, there are about 4,500 cities with a population of at least 100,000. If you had the power to do such a thing, you could destroy every city with 3 bombs and still have 1,500 left over. This would kill more than 3 billion people. Crazy, right? Well, let us try grouping them together! If we placed them in the Amazon rainforest and exploded them all at once, here is what would happen. The explosion would carry the same energy as 3,000,000,000 tons of TNT: 3 billion tons! That is equivalent to 15 deadly volcanic eruptions combined, or enough TNT to build a complete city.

You push the button, and 3 billion tons of TNT equivalent goes boom! The blast vaporizes everything within 50 km, sends a blast wave across $3,000 \text{ km}^2$, and ignites everything within 250 km. If you are on Earth, you can hear it explode multiple times as the sound circles the globe. A mushroom cloud rises into the sky and touches outer space while rocks of all sizes shoot into orbit and fall back down. After it settles, a 10 km wide crater forms while forests continue to burn. Then comes the worst part: radioactive material spreads across the Earth, increasing cancer rates and killing wildlife. Human life would survive if we made it through decades of a cold radioactive winter, but the planet would need hundreds, if not thousands, of years to recover.

So, do you still want to press the red button?

